# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

# CHEMISTRY

Past Paper 2024 • Paper Code: 6485 / 6488

Retyped faithfully from the original board paper — same questions, no changes.

## GROUP – I

### Paper: I Objective (iii)

کیمسٹری

**Time Allowed:** 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6485

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

نوٹ:- ہر سوال کے چار ممکنہ جوابات A, B, C اور D دیئے گئے ہیں۔ جوابی کاپی پر ہر سوال کے سامنے دیئے گئے دائروں میں سے درست جواب کے مطابق دائرہ کو مار کر یا پین سے پُر کیجیے۔ ایک سے زیادہ دائروں کو پُر کرنے یا کاٹ کر پُر کرنے کی صورت میں مذکورہ جواب غلط تصور ہوگا۔

Q1. Orbitals having same energy are called:

(A) Degenerate orbitals
(B) Valence orbitals
(C) d-orbitals
(D) s and p orbitals

Q2. Amorphous solids:

(A) Can possess small regions of orderly arrangement of atoms
(B) Have sharp melting point
(C) Have perfect arrangement of atoms
(D) Undergo clean cleavage when cut with knife

Q3. Halogen that exists as solid at room temperature is:

(A) $F_2$
(B) $Cl_2$
(C) $Br_2$
(D) $I_2$

Q4. The order of rate of diffusion of gases $NH_3$, $SO_2$, $Cl_2$ and $CO_2$ is:

(A) $NH_3 > SO_2 > Cl_2 > CO_2$
(B) $Cl_2 > SO_2 > CO_2 > NH_3$
(C) $NH_3 > CO_2 > Cl_2 > SO_2$
(D) $NH_3 > CO_2 > SO_2 > Cl_2$

Q5. What is the most abundant form of matter around us on our earth.

(A) Gas
(B) Liquid
(C) Solid
(D) Plasma

Q6. How many steps are involved in complete quantitative characterization?

(A) 2
(B) 3
(C) 4
(D) 5

Q7. Which of the given is used as decolourizing agent in crystallization?

(A) Graphite
(B) Animal Charcoal
(C) $H_2SO_4$
(D) KOH

Q8. Which of the given is a mono isotopic element?

(A) Fluorine
(B) Chlorine
(C) Silver
(D) Calcium

Q9. The largest number of molecules are present in:

(A) 4.8g of $C_2H_5OH$
(B) 3.6g of $H_2O$
(C) 2.8g of CO
(D) 5.4g of NO

Q10. If salt bridge is not used between two half cells then the voltage:

(A) Decreases slowly
(B) Drops to zero
(C) Decreases rapidly
(D) Does not change

Q11. The rate of reaction:

(A) Decreases as the reaction proceeds
(B) Increases as the reaction proceeds
(C) Remains the same as the reaction proceeds
(D) May decrease or increase as the reaction proceeds

Q12. A solution of glucose is 10 % w/v. The volume in which its one g.mole is dissolved will be:

(A) 1 $dm^3$
(B) 1.8 $dm^3$
(C) 200 $cm^3$
(D) 900 $cm^3$

Q13. Which of the given is weak acid?

(A) HCl
(B) $H_2SO_4$
(C) $CH_3COOH$
(D) $HNO_3$

Q14. One calorie is equal to:

(A) 0.4184 J
(B) 41.84 J
(C) 418.4 J
(D) 4.184 J

Q15. Which of the given species has unpaired electrons in antibonding molecular orbitals?

(A) $O_2^{2-}$
(B) $B_2$
(C) $N_2^{2-}$
(D) $F_2$

Q16. Molecular shape of $SO_3$ according to VSEPR Theory:

(A) Triangular Planner
(B) Linear
(C) Pyramidal
(D) Tetrahedral

Q17. Visible range contains wave length in between:

(A) 200-400 n.m
(B) 400-750 n.m
(C) 200-800 n.m
(D) 800-1200 n.m

## Paper: I Subjective

کیمسٹری

**Time Allowed:** 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any THREE questions from Section C.

نوٹ: حصہ دوم لازمی ہے۔ حصہ سوئم میں سے کوئی سے تین سوالات کے جوابات تحریر کیجیے۔

Q2. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Why 23 g of sodium and 238 g of uranium have equal number of atoms in them?

(ii) Define gram formula giving one example.

(iii) Write down two characteristics of plasma.

(iv) What do you mean by partition chromatography? Give Example.

(v) Define sublimation and name at least two sublimed solids.

(vi) How can rate of filtration be increased by flutted filter paper?

(vii) How does values of equilibrium constant (Kc) help predict extent of a reaction?

(viii) Why lighter gases diffuse more rapidly than heavier ones?

(ix) How are acidic and basic buffers prepared? Give one example in each case.

(x) State Charle's law of gases. Give its mathematical form.

(xi) Differentiate between reversible and irreversible reactions with examples.

(xii) How Mg-atom is twice heavier than that of carbon atom? Explain.

Q3. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Boiling needs constant supply of heat. Give reason.

(ii) How chloroform and acetone are miscible with each other?

(iii) How does liquid crystals act as temperature sensors?

(iv) Why Molecular solids are soft and easily compressible?

(v) What is the reason for production of positive rays?

(vi) Differentiate between Zeeman and Stark effect.

(vii) Give two points for the significance of Moseley's law.

(viii) Why boiling points of solvents increase due to the presence of solutes?

(ix) What do you mean by Heterogeneous catalysis? Give two examples.

(x) Draw the shapes of 's' and 'p' orbitals.

(xi) Rate of a reaction is everchanging parameter under the given conditions. Justify it.

(xii) Define "Hydrolysis". Give two examples.

Q4. (Marks 12) Write short answers to any SIX parts. (6x2=12)

(i) Write down the electrode reactions in alkaline battery.

(ii) Define dipole moment. Give its various units.

(iii) What is pressure-volume work?

(iv) Why do the ionization energies of elements decrease down the group of periodic table although the nuclear charge increases?

(v) The abnormality of bond length in HI is less prominent than that of HCl. Give the reason.

(vi) Why is MOT superior to VBT?

(vii) Define standard enthalpy of a reaction. Give an example.

(viii) Differentiate between exothermic and endothermic reactions.

(ix) SHE acts as anode when connected with Cu electrode but as cathode when connected with Zn electrode. Give the reason.

> Note: Attempt any THREE questions. Each question carries EIGHT (08) marks. (8x3=24)

Q5. (Marks 8)

(a) [4] What are ions? Give their different types. Under what conditions these can be generated?

(b) [4] Name the factors affecting the "London Forces". Explain the boiling points of halogens in detail.

Q6. (Marks 8)

(a) [4] Calculate the density of $CH_4(g)$ at 0°C and 1 atm pressure.

(b) [4] How e/m value of electron is measured?

Q7. (Marks 8)

(a) [4] Define and explain co-ordinate covalent bond with three examples.

(b) [4] What is the percentage ionization of acetic acid in a solution in which 0.1 moles of it has been dissolved per $dm^3$ of the solution (Ka of acetic acid=$1.85x10^{-5}$)?

Q8. (Marks 8)

(a) [4] Describe in detail the measurement of enthalpy of combustion with the help of Bomb Calorimeter with diagram.

(b) [4] Define primary storage cell. Explain silver oxide battery in detail.

Q9. (Marks 8)

(a) [4] What is Raoult's Law? Explain it with three statements also.

(b) [4] Define order of reaction. Name various methods for its determination and explain only half life method in detail.

## GROUP – II

### Paper: I Objective (iv)

کیمسٹری

| **Time Allowed:** 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6488 |
|---|---|---|

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

نوٹ:- ہر سوال کے چار ممکنہ جوابات A, B, C اور D دیئے گئے ہیں۔ جو ابی کاپی پر ہر سوال کے سامنے دیئے گئے دائروں میں سے درست جواب کے مطابق دائرہ کو مار کر یا پین سے پُر کیجیے۔ ایک سے زیادہ دائروں کو پُر کرنے یا کاٹ کر پُر کرنے کی صورت میں مذکورہ جواب غلط تصور ہوگا۔

Q1. The quantum number which gives information about degeneracy of orbitals in space is:

(A) Principal quantum number
(B) Azimuthal quantum number
(C) Magnetic quantum number
(D) Spin quantum number

Q2. The molecules of $CO_2$ in dry ice form:

(A) Ionic crystals
(B) Covalent solids
(C) Molecular solids
(D) Metallic solids

Q3. Acetone and Chloroform are soluble in each other due to:

(A) Hydrogen bonding
(B) Ion-dipole forces
(C) Instantaneous dipole
(D) London dispersion forces

Q4. The partial pressure of oxygen in the air is:

(A) 116 torr
(B) 200 torr
(C) 159 torr
(D) 150 torr

Q5. The drying agent used in vacuum desiccator is:

(A) Benzoic acid
(B) Glucose
(C) Silical gel
(D) Animal charcoal

Q6. Solvent extraction is an equilibrium process and is controlled by:

(A) Law of mass action
(B) The amount of solvent
(C) Distribution law
(D) The amount of solute

Q7. The molar volume of $CO_2$ is maximum at:

(A) S.T.P
(B) 127°C and 1 atm
(C) 0°C and 2 atm
(D) 273°C and 2atm

Q8. The number of moles of $CO_2$ which contain 8.0g of oxygen is:

(A) 0.25
(B) 0.50
(C) 1.0
(D) 1.50

Q9. The empirical formula of glucose is:

(A) CHO
(B) $C_6H_{12}O_6$
(C) $C_2HO$
(D) $CH_2O$

Q10. The reduction potential of Zinc electrode is:

(A) 0.76 volt
(B) −0.76 volt
(C) −0.34 volt
(D) 0.34 volt

Q11. The catalyst used in the decomposition of $KClO_3$ is:

(A) $CuCl_2$
(B) $V_2O_5$
(C) $MnO_2$
(D) NO

Q12. The molal boiling point constant is the ratio of the elevation in boiling point to:

(A) Molarity
(B) Mole fraction of solvent
(C) Molality
(D) Mole fraction of solute

Q13. The pH of human blood is maintained at:

(A) 7.0
(B) 7.35
(C) 4.0
(D) 14.0

Q14. For a given process, the heat changes at constant pressure (qp) and at constant volume (qv) are related to each other as:

(A) qp=qv
(B) qp<qv
(C) qp>qv
(D) qp=qv/2

Q15. Which of the hydrogen halides has the highest percentage of ionic character?

(A) HCl
(B) HBr
(C) HF
(D) HI

Q16. $C_2H_4$ (ethene) shows hybridization:

(A) $sp^3$
(B) $sp^2$
(C) sp
(D) $dsp^2$

Q17. The wave number of the light emitted by a certain source is $2x10^6$ $m^{-1}$. The wavelength of this light will be:

(A) 500 nm
(B) 500 m
(C) 200 nm
(D) $5x10^7$m

## Paper: I Subjective

کیمسٹری

**Time Allowed:** 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any THREE questions from Section C.

نوٹ: حصہ دوم لازمی ہے۔ حصہ سوئم میں سے کوئی سے تین سوالات کے جوابات تحریر کیجیے۔

Q2. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Define Avogadro's number with a suitable example.

(ii) Write down two assumptions of stoichiometry.

(iii) Many chemical reactions involve limiting reactant as taking place in our surrounding. Justify.

(iv) Why is crystallization a better technique for separation and purification?

(v) Name any four sublimed solids.

(vi) What is Rf value? Also write down its formula.

(vii) Deduce Boyle's law with the help of Kinetic theory of gases.

(viii) Write down any four applications of plasma.

(ix) How does buffer act?

(x) The plot of PV versus P is a straight line at constant temperature and with a fix number of moles of an ideal gas. Explain.

(xi) How does equilibrium constant (Kc) predict direction of a reaction?

(xii) Give optimum conditions to get maximum yield of Ammonia ($NH_3$).

Q3. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) What are liquid crystals? Write down their any two uses.

(ii) Evaporation causes cooling, why?

(iii) Define Anisotropy. Give example.

(iv) Ionic crystals are highly brittle, why?

(v) Write down any two properties of neutrons.

(vi) What is stark effect?

(vii) What is meant by dual nature of matter?

(viii) Define spin quantum number.

(ix) What is percentage weight / weight? Give example.

(x) Non ideal solutions do not obey Roults' law, why?

(xi) What is heterogeneous catalysis? Give example.

(xii) Define energy of activation.

Q4. (Marks 12) Write short answers to any SIX parts. (6x2=12)

(i) Why atomic radius cannot be measured precisely?

(ii) Size of an anion is always larger than its parent atom. Justify.

(iii) Why is the second electron affinity of oxygen positive?

(iv) Why $He_2$ does not exist under normal conditions.

(v) Why is it necessary to mention physical states of reactants and products in thermo chemical equation?

(vi) Define internal energy with one example.

(vii) Calculate oxidation number of Cr in $K_2Cr_2O_7$.

(viii) Write down importance of standard hydrogen electrode.

(ix) Define heat and work.

> Note: Attempt any THREE questions. Each question carries EIGHT (08) marks. (8x3=24)

Q5. (Marks 8)

(a) [4] What is meant by combustion analysis? Draw neat diagram. Also write down formulas to calculate percentages of carbon and hydrogen.

(b) [4] What are ionic solids? Write down any six properties of ionic solids.

Q6. (Marks 8)

(a) [4] Calculate the mass of 1 $dm^3$ of $NH_3$ gas at 30°C and 1000 mmHg pressure considering that $NH_3$ is behaving ideally.

(b) [4] Write down any eight properties of cathode rays.

Q7. (Marks 8)

(a) [4] Write down four postulates of VSEPR theory.

(b) [4] The solubility product of $Ag_2CrO_4$ is $2.6x10^{-2}$ at 25°C. Calculate the solubility of the compound.

Q8. (Marks 8)

(a) [4] State first law of thermodynamics. How will you prove that work done by the system is negative ($W=-P\Delta V$).

(b) [4] What is electrochemical series? Explain the following application in detail; (i) Comparison of the relative tendency of metals and non metals to get oxidized or reduced.

Q9. (Marks 8)

(a) [4] Explain Beckmann method for the measurement of freezing point depression with diagram.

(b) [4] Define order of reaction. Give one example of first order, second order and third order reaction.